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RAFFLES JUNIOR COLLEGE
Preliminary Examination

H2 BIOLOGY
PAPER 1

9747/1
18th September 2008

1 hour 15 min

Additional material: Optical Mark Reader sheet

READ THESE INSTRUCTIONS FIRST

1. Answer **all** questions.
2. Each question has 4 options. Choose the one you consider most appropriate and record your choice on the OMR sheet provided.
3. Each correct answer will score half a mark. Marks will not be deducted for wrong answers.
4. Use **only** a soft pencil to shade the **Optical Mark Reader (OMR)** sheet.
5. Do not bend or fold the OMR sheet. Erase all mistakes completely.
6. Hand in your OMR sheet.

This document consists of **15** printed pages.

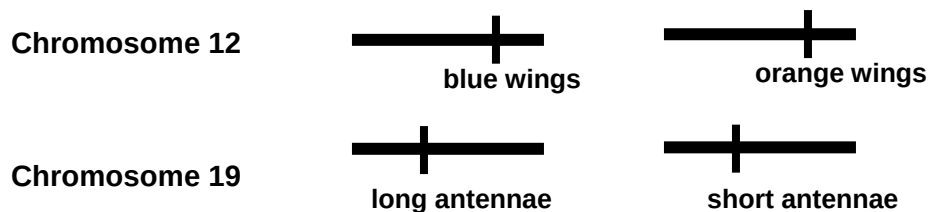


Raffles Junior College
Internal Examination

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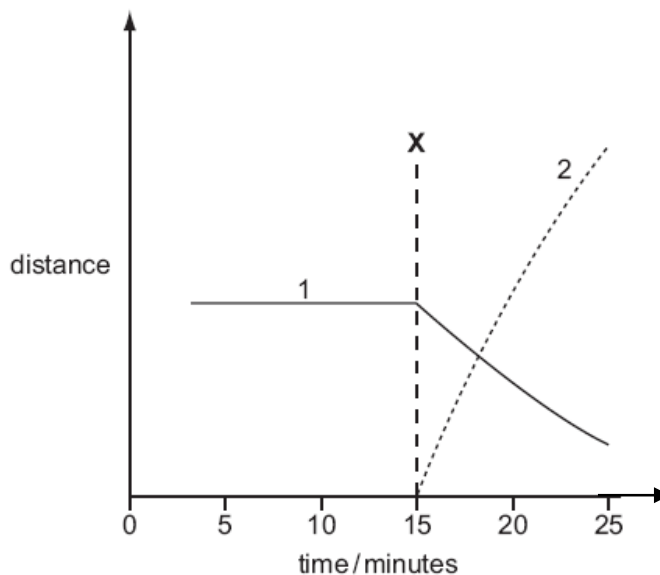
1. The enzyme amylase hydrolyses starch into disaccharides. Amylase is unable to breakdown cellulose because
- A cellulose does not contain glycosidic bonds.
 - B starch consists of glucose; cellulose consists of other monosaccharides.
 - C the bonds between sugars in cellulose are much stronger.
 - D the sugars in cellulose are bonded together differently than in starch, hence cellulose have a different shape.
2. Which of the following does not describe the active site of an enzyme?
- A Its specificity is defined by the arrangement of certain atoms.
 - B It is usually a crevice or cleft.
 - C It initially binds substrates by weak attractions.
 - D It is two-dimensional in structure.
3. Identify whether the following organelles are commonly found in animal cells, plant cells, or both animal and plant cells.
- i. chloroplasts
 - ii. mitochondria
 - iii. large vacuole
 - iv. ribosome
 - v. nucleolus
- a. animal cells
 - b. plant cells
 - c. both animal and plant cells
- A i = b ; ii = c ; iii = c ; iv = c ; v = c
 - B i = a ; ii = b ; iii = a ; iv = c ; v = a
 - C i = c ; ii = c ; iii = b ; iv = a ; v = b
 - D i = b ; ii = c ; iii = b ; iv = c ; v = c

4. A certain (hypothetical) organism is diploid, has either blue or orange wings as the consequence of one of its genes, and has either long or short antennae as the result of a second gene, as shown below.



If an organism is heterozygous for both genes, what will be the possible chromosomes and alleles found in one of her eggs?

- A Two chromosomes 12 with blue and orange wings alleles.
 B Chromosome 12 with an orange wings allele and chromosome 19 with long antennae allele.
 C Chromosome 19 with both orange wings allele and short antennae allele due to crossing over.
 D Chromosome 12 with blue and orange wings alleles and chromosome 19 with long and short antennae alleles.
5. The graph shows measurements during one mitotic cell cycle. Which stage of mitosis commences at X and which measurements are illustrated by curves 1 and 2?



	Stage beginning at X	Distance between centromeres of chromosomes and poles of spindle	Distance between the centromeres of sister chromatids
A	Anaphase	1	2
B	Anaphase	2	1
C	Metaphase	1	2
D	Metaphase	2	1

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6. Which one of the following is associated with accuracy of DNA replication?
- A High energy bonds between phosphate groups.
 - B Covalent bonds between bases and phosphate groups.
 - C Hydrogen bonds between bases.
 - D Covalent bonds between bases.
7. Assume that the average amino acid residue has a molecular weight of 110. The DNA strand coding for a polypeptide chain of molecular weight 20,000 has a length of
- A 182 nucleotides.
 - B 252 nucleotides.
 - C 540 nucleotides.
 - D 760 nucleotides.
8. What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?
- I. The anti-parallel arrangement of the DNA strands.
 - II. The RNA primers are required to initiate DNA elongation.
 - III. DNA polymerase joins new nucleotides to the 3' end of the growing strand.
 - IV. Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.
- A II and IV
 - B I and III
 - C I and IV
 - D II and III
9. Genetic drugs are short sequences of nucleotides. One example is an anti-sense drug made from RNA nucleotides that are complementary to the protein-coding mRNA.
- Which is the process that is affected by the anti-sense drug and what is its most probable mode of action?
- A It prevents DNA replication by binding to the anti-sense strand of DNA that codes for the disease-causing protein.
 - B It prevents transcription by binding to the sense strand of DNA that codes for the disease-causing protein.
 - C It prevents translation by binding to the mRNA that contains information for the disease-causing protein.
 - D It prevents the normal functioning of disease-related protein when it is translated into a protein that inhibits the disease-causing protein.

10. During a viral infection, attachment is usually specific to a particular cell type because
- A the virus is attracted to the appropriate host cells by proteins secreted into the extracellular fluid.
 - B the virus recognises and binds to specific molecules in the cytoplasm of the host cell.
 - C the virus recognises and binds to specific molecules on the surface of the host cell.
 - D the host cell produces channel proteins that provide passageways for the viruses to enter the cytoplasm.
11. Genetic diversity is maintained by all of the following except
- A mutation.
 - B transformation.
 - C transduction
 - D binary fission.
12. Which of the following is not true for both prokaryotic and eukaryotic genomes?
- A They both have extrachromosomal DNA.
 - B They both have non-coding regions.
 - C They both have regulatory sequences.
 - D They are both associated with histones.
13. Gene amplification is a mechanism known to result in
- I synthesis of large amounts of ribosomes.
 - II many copies of the same mRNA coding for drug resistance.
 - III an increased number of gene clusters.

Which of the following are correct?

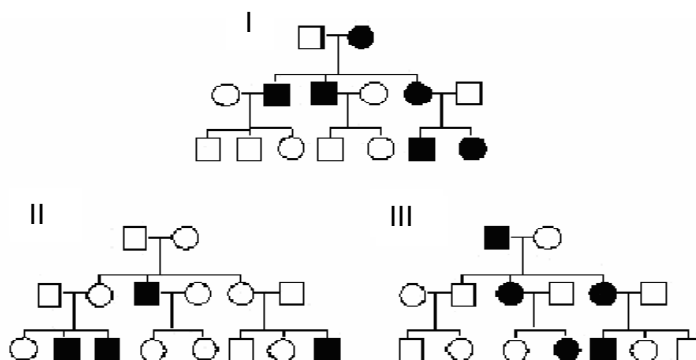
- A I and III only
 - B II and III only
 - C I and II only
 - D I, II and III
14. Which of the following conditions must be present for cancer formation?

	Number of alleles mutated for each proto-oncogene	Number of alleles mutated for each tumour-suppressor gene
A	1	1
B	2	1
C	1	2
D	2	2

15. In sweet-pea plants, flowers may be either purple or white. Colour is determined by two allelomorphous pairs of genes (A, a and B, b). If at least one dominant gene from each allelomorphous pair is present, then the colour is purple. All other genotypes are white. If a purple plant with a genotype AaBb had two pure-breeding white parents, then the genotypes of those parents have been

A AAbb and aabb
 B AAbb and AAbb
 C aaBB and AAbb
 D Aabb and aaBb

16. A man is diagnosed with a glucose 6-phosphate deficiency subsequent to taking primaquine for malaria due to *Plasmodium vivax*. Given that glucose 6-phosphate deficiency is a sex-linked recessive trait, which of the following pedigrees best illustrates the inheritance pattern of this trait?



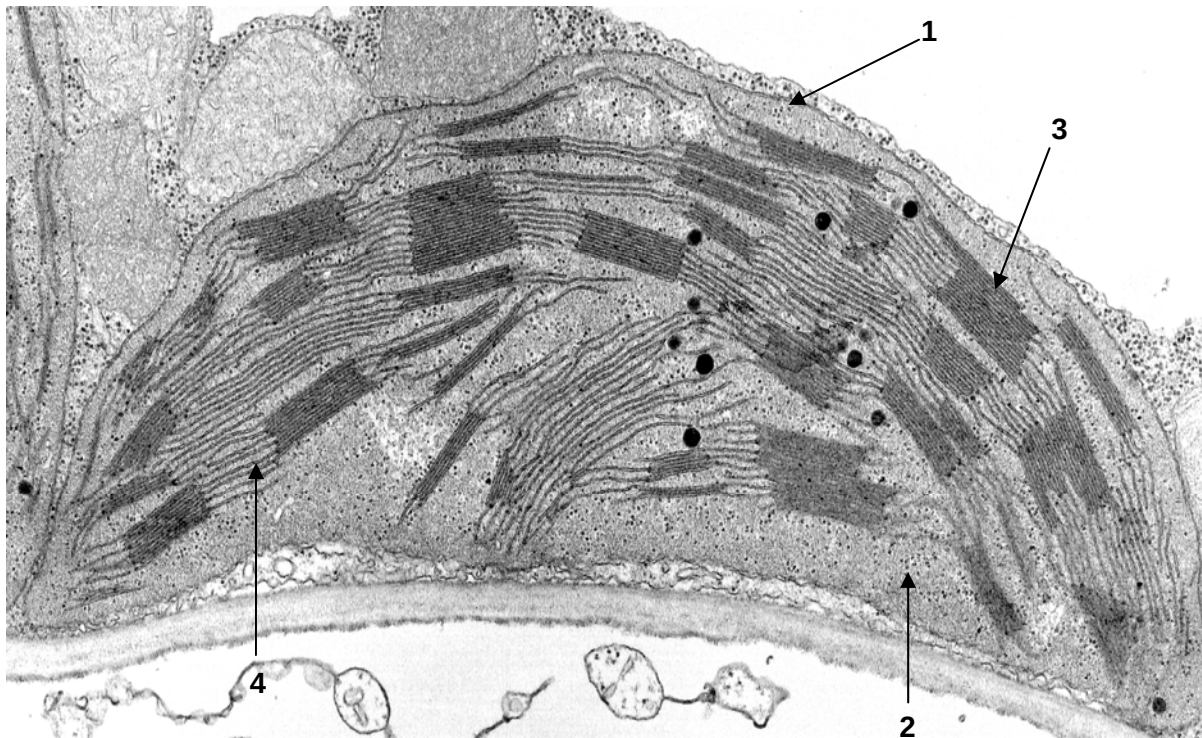
A I only
 B II only
 C I and III only
 D II and III only

17. Marian's father is colourblind, as is her maternal grandfather (her mother's father). Marian herself has normal colour vision. Marian and her husband, Martin, who is also colourblind, have just had their first child, a son they have named Mickey.

What is the probability that this child will have colourblindness?

A 1/8
 B 1/4
 C 1/2
 D 1

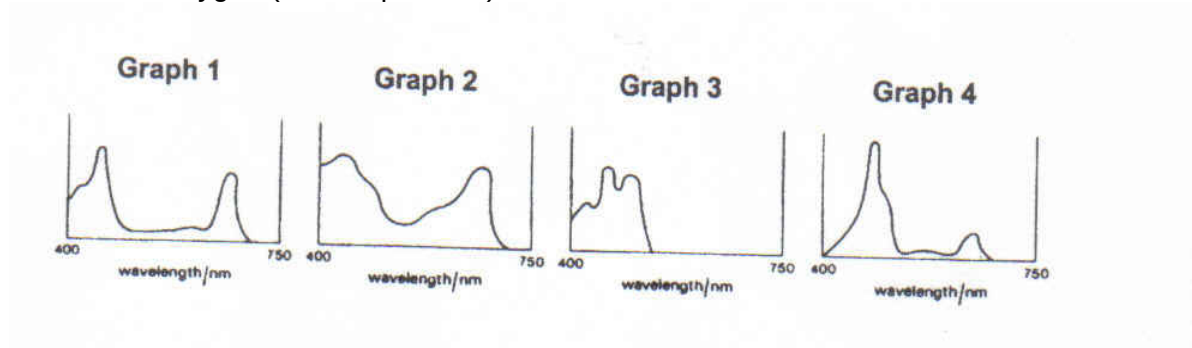
18. Which of the following is correctly associated with the function or characteristic of the labelled structure?



	help to maintain high H^+ concentration	made up of phospholipid	has stalked particles embedded in it	where ATP is synthesised
A	1 and 2	1, 2, 3 and 4	4	2
B	3 and 4	2 and 4	2 and 3	1 and 2
C	1 and 4	1, 2 and 3	1 and 4	2 and 4
D	3 and 4	1, 3 and 4	3 and 4	2

19. The four graphs below show the absorption spectra of photosynthetic pigments and the action spectrum of photosynthesis for a plant containing those pigments.

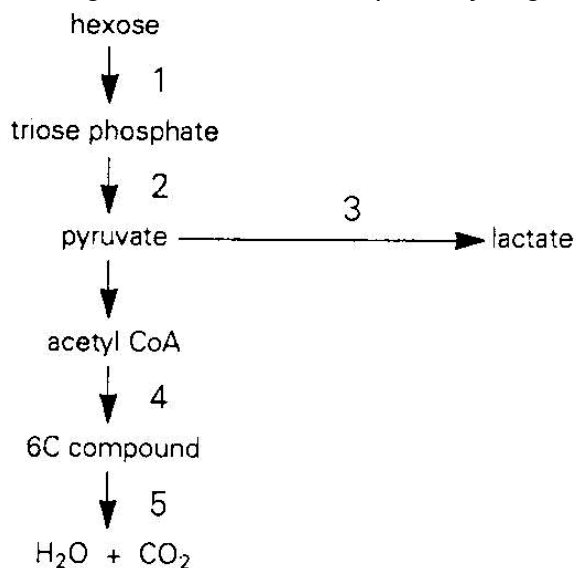
The x-axes show wavelength in nanometres (nm). The y-axes show either the percentage of light absorbed (absorption spectra) or the rate of photosynthesis measured by the rate of evolution of oxygen (action spectrum)



Which one of the following series identifies the four graphs?

	absorption spectra			action spectrum
	chlorophyll a	chlorophyll b	carotenoids	
A	4	2	1	3
B	3	1	2	4
C	1	4	3	2
D	2	3	4	1

20. The diagram summarises the pathway of glucose breakdown.



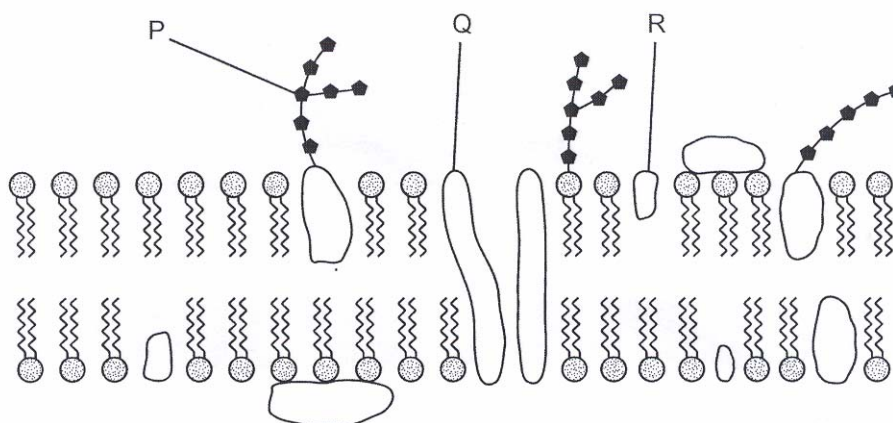
Which two steps result in a net increase of ATP?

- A 1 and 3
 B 2 and 3
 C 2 and 5
 D 4 and 5

21. Each time a molecule of glucose is completely oxidised via aerobic respiration, how many oxygen (O_2) molecules are required?

- A 1
- B 2
- C 6
- D 12

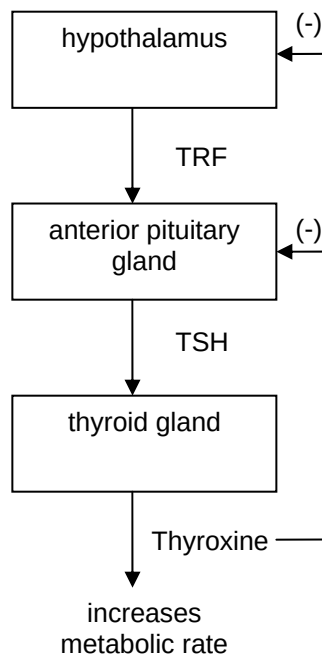
22. The diagram below shows part of a cell surface membrane.



What is the correct function for each of the structures labelled?

	regulates membrane fluidity	forms hydrogen bonds with water to stabilise membrane	transports ions and large polar molecules
A	R	R	Q
B	P	Q	R
C	Q	R	P
D	R	P	Q

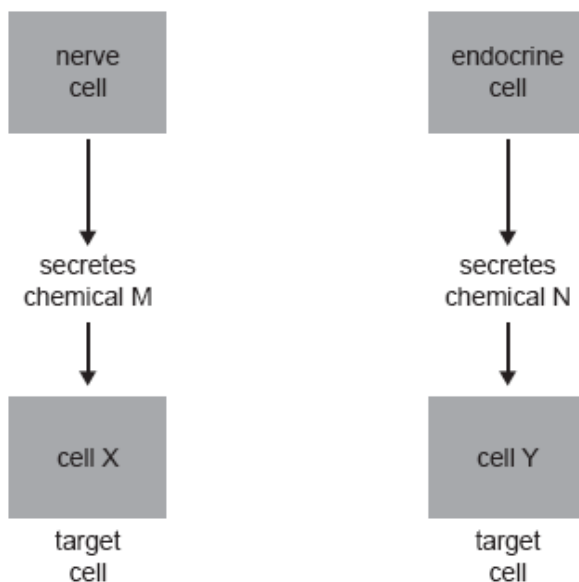
23. Homeostasis often involves multi-step pathways. An example is given in the diagram below, which shows the hormonal feedback mechanism that controls body temperature. [TRF = thyrotrophin-releasing factor ; TSH = thyroid-stimulating hormone]



A rise in temperature causes a decrease in the amount of TRF released by the hypothalamus. As a result of this, the level of thyroxine will

- A rise with a consequent fall in the basic metabolic rate.
- B fall with a consequent fall in the basic metabolic rate.
- C inhibits production of TRF.
- D inhibits production of TSH.

24. The diagram shows how a nerve cell and an endocrine cell interact with cell X and cell Y respectively.



Which of the following is true of both interactions?

- A Diffusion is involved in the movement of chemicals M and N to their respective target cells.
 - B Chemical M is secreted faster than chemical N.
 - C When both cells X and Y are stimulated, phosphorylation cascades must be initiated.
 - D Both chemicals M and N must bind to cell-surface receptors on cell X and Y respectively.
25. All of the following statements about events involved in glycogen metabolism are true except
- A cAMP-activated protein kinase stimulates glycogen synthesis.
 - B kinases are activated by phosphorylation.
 - C cAMP levels are raised during glycogen hydrolysis.
 - D receptors activated by cross-phosphorylation leads to glycogenesis.

26. Tetrodotoxin, a puffer fish toxin, blocks voltage-gated sodium channels. Black widow spider's venom causes the voltage-gated calcium channels to be constantly open. Crotoxin binds irreversibly to acetylcholine receptors. What will happen to the nerve transmission if each toxin is applied?

	tetrodotoxin	black widow spider's venom	crotoxin
A	block action potentials along axon	reduce transmission of impulse across synapse	increase transmission of impulse across synapse
B	increase transmission of impulse across synapse	reduce transmission of impulse across synapse	block action potentials along axon
C	block action potentials along axon	increase transmission of impulse across synapse	reduce transmission of impulse across synapse
D	reduce transmission of impulse across synapse	block action potentials along axon	increase transmission of impulse across synapse

27. During an action potential, how is the resting potential restored?

- A by an increase in the membrane's permeability to potassium and chloride ions
- B by the opening of sodium activation gates
- C by the action of the sodium-potassium pump
- D by the opening of voltage-sensitive potassium channels and the closing of sodium activation gates

28. A drug might act as a stimulant of the nervous system if it

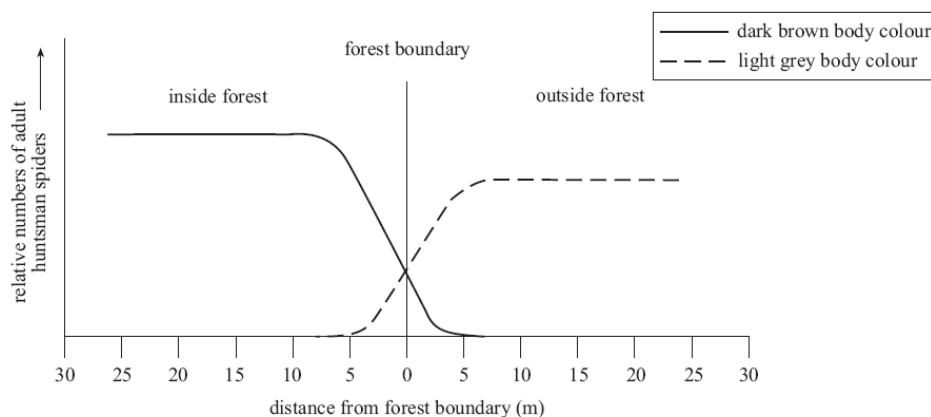
- A increases the sensitivity of the postsynaptic membrane to acetylcholine.
- B stimulates the activity of acetylcholinesterase in the synaptic cleft.
- C increases the sensitivity of the presynaptic membrane to acetylcholine.
- D increases the release of substances that cause the hyperpolarization of the neurons.

29. Given a population that contains genetic variation, what is the correct sequence of the following events, under the influence of natural selection?

1. Differential reproduction occurs.
2. A new selective pressure occurs.
3. Allele frequencies within the population changes.
4. Environmental change occurs.

- A 2, 4, 1, 3
- B 4, 2, 1, 3
- C 4, 1, 2, 3
- D 2, 4, 3, 1

30. The graph below shows the distribution of huntsman spiders at a forest boundary.



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that

- A the two populations of spiders were once different species.
 - B the two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.
 - C the differences in the two habitats had changed the physical appearance of individual spiders.
 - D a particular body colour provided a selective advantage to spiders in a particular habitat.
31. Allopatric speciation should involve
- A genetic drift.
 - B disruptive selection.
 - C geographic isolation.
 - D none of the above.
32. A method used to detect a particular DNA sequence within a mixture of many DNA fragments is
- A DNA sequencing.
 - B DNA fingerprinting.
 - C colony hybridisation.
 - D Southern blotting.

33. To separate polypeptides of different sizes, they must be subjected to electrophoresis
- A so that they migrate from the wells to the negative electrode.
 - B after digestion with enzyme.
 - C after PCR amplification.
 - D under denaturing conditions.
34. You are attempting to clone a gene of interest into the *lacZ* gene of a plasmid containing an ampicillin-resistance gene. The recombinant molecule was then introduced into a bacterial host sensitive (i.e. not resistant) to ampicillin. You grow your potential transformants on media containing both ampicillin and X-gal. Which of the following results would most likely indicate a successful clone?
- A cells which do not grow on the ampicillin plates
 - B cells which form white colonies on the ampicillin plates
 - C cells which form blue colonies on the ampicillin plates
 - D cells which form light blue (half blue, half white) colonies on the ampicillin plates
35. The sum total of proteins produced by an organism is called its proteome. Which of the following statements regarding the analysis of proteomes (a field called proteomics) is false?
- A Alternative splicing can explain why a proteome may be larger than the total number of genes in a genome.
 - B The field of proteomics aims to describe how phenotypes are affected by the expression of proteins.
 - C Proteomics can reveal differences between genetically related species.
 - D Proteomics has proven the 'one gene, one polypeptide' hypothesis.
36. What are some similarities between all cancer cells and all stem cells?
- A They replicate indefinitely, lack cell-cell adhesion and are able to move from one location in the body to another.
 - B They lack contact inhibition, are non-differentiated and are regulated by molecular signals.
 - C They are able to move from one location in the body to another, are regulated by molecular signals and can be found in various parts of the body.
 - D They replicate indefinitely, are non-differentiated and can be found in various parts of the body.

37. Gene therapy is used for the treatment of

1. cystic fibrosis and
2. severe combined immunodeficiency (SCID).

The difference between them is

- A 1 is a transient treatment while 2 can possibly be a permanent treatment.
- B 1 usually uses the *ex vivo* approach while 2 usually uses the *in vivo* approach.
- C 1 usually uses adenoviruses as vectors while 2 usually uses liposomes as vectors.
- D 1 treats a recessive disorder while 2 treats a dominant disorder.

38. What are some key reasons explaining why gene therapy is not (yet) an effective treatment for genetic diseases?

- I lack of knowledge of DNA sequences
- II lack of efficient and safe gene delivery systems
- III cost of the technology
- IV some diseases involve multiple genes
- V problems with appropriate gene regulation

- A II only
- B I, III and IV only
- C II, IV and V only
- D All of the above

39. Increased genetic diversity following extended time in a plant tissue culture is a problem called

- A gene alteration.
- B culture shock.
- C acclimatisation shock.
- D somaclonal (somatic) variation.

40. In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?

- A Chloroplasts are surrounded by a double membrane.
- B There are no chloroplasts in pollen of most plant species.
- C Chloroplasts are smaller than the nucleus.
- D There is no DNA in chloroplasts.